PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

Academic Year: 2026-2027
Semester: V
Class / Branch: TE B
Subject: Agile Software Development & DevOps
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Date of Performance: 20/10/2026
Date of submit: 20/10/2026

Level Up Assignment

Q1 : Implement the Scrum framework for a selected software development project (e.g., Library Management System / College Event Management System) using Jira by performing the following activities:

1. Create a Scrum project in Jira and define the Scrum team roles (Product Owner, Scrum Master, and Development Team).
2. Identify project requirements and create User Stories with suitable Acceptance Criteria.
3. Create and prioritize the Product Backlog based on business requirements.
4. Conduct Sprint Planning by selecting User Stories and preparing a Sprint Backlog with task assignments and timelines.
5. Simulate Scrum activities by updating task status, conducting Daily Scrum discussions, and tracking sprint progress using a Scrum Board.
6. Demonstrate Sprint Review and Retrospective activities and prepare suitable Scrum artifacts, including the Product Backlog, Sprint Backlog, and Increment.

Implementation:

The College ERP System (student academic portal) was selected as the project. Scrum was implemented in the Jira Scrum project College ERP System (project key SCRUM). The Scrum events were simulated by the student for this academic exercise, and the working increment is a browser-based prototype stored in Level-Up-Assignment/Q1-Scrum-Jira/Increment in the repository.

1. Scrum project and team roles

Role	Responsibility in the project	Played by
Product Owner	Owns the Product Backlog, writes and prioritises user stories, defines acceptance criteria and clarifies requirements	Vighnesh Pingale (simulated role)
Scrum Master	Facilitates Sprint Planning, Daily Scrum, Review and Retrospective, removes impediments and keeps the team following Scrum	Vighnesh Pingale (simulated role)
Development Team	Plans, builds and tests the work selected for the sprint	Vighnesh Pingale (simulated role)

2. Requirements, user stories and acceptance criteria

Four user stories were created in Jira for the core student features. Acceptance criteria were written in the description of each story.

Key	User story	Priority
SCRUM-17	Student Login to College ERP	Highest
SCRUM-18	View Academic Dashboard: as a student, I want to view my academic dashboard so that I can quickly access my attendance, examination results, and course information	High

Key	User story	Priority
SCRUM-19	View Student Attendance: as a student, I want to view my attendance records so that I can monitor my attendance percentage for each subject	High
SCRUM-20	View Examination Results: as a student, I want to view my examination results so that I can review my academic performance	High

Acceptance criteria (SCRUM-18): an authenticated student can open the dashboard; it displays links to attendance, examination results and course information; it shows the information of the logged-in student; an appropriate message is shown when information is unavailable; unauthenticated users cannot access the dashboard.

Acceptance criteria (SCRUM-19): an authenticated student can open the attendance section; attendance records are displayed for each enrolled subject; the attendance percentage is calculated and displayed for each subject; the student can identify subjects where attendance is below the required threshold.

Acceptance criteria (SCRUM-20): an authenticated student can open the examination-results section; results are displayed for the logged-in student; each result shows the subject, examination name, marks and result status when available; a student cannot view another student's results.

3. Product Backlog and prioritisation

The Product Backlog was ordered by business value. Student login is the entry point to every other feature and therefore has the Highest priority; the dashboard, attendance and examination results are High priority and follow it.

4. Sprint Planning and Sprint Backlog

SCRUM Sprint 2 was created with a two-week timeline (2 Nov – 16 Nov) and the four user stories were moved from the backlog into the sprint. Each story was broken into a planning sub-task and assigned in Jira, forming the Sprint Backlog.

Sub-task	Parent story	Task	Priority
SCRUM-21	SCRUM-17	Plan login validation and authentication tests: login form requirements, validation of registered users, error handling for invalid credentials, and test cases for successful login, failed login and logout	High
SCRUM-22	SCRUM-18	Plan academic dashboard and navigation: layout, links to attendance, results and course information, identification of the logged-in student and handling of unavailable information	High
SCRUM-23	SCRUM-19	Plan attendance display and percentage validation: records per subject, percentage calculation, low-attendance identification and missing records	High
SCRUM-24	SCRUM-20	Plan examination results display and access checks: subject, examination, marks and status, unpublished results and restricted access	High

5. Scrum activities: task status, Daily Scrum and Scrum Board

Task status was updated on the Scrum Board (To Do, In Progress, Done). A Daily Scrum was simulated and recorded as a comment on the sprint work item, following the standard three questions with an action and a progress note:

Daily Scrum – Day 1 (academic simulation)	
Yesterday	The team finalised the Student Login user story and its acceptance criteria.

Daily Scrum – Day 1 (academic simulation)	
Today	The team will prepare test cases for valid credentials, invalid credentials and secure logout.
Blocker	The password validation rules and error-message requirements need to be clarified.
Action	The Product Owner will clarify the login requirements and the Development Team will refine the test cases accordingly.
Progress	Login test planning has started. Application implementation has not yet been demonstrated.

6. Sprint Review, Retrospective and Scrum artifacts

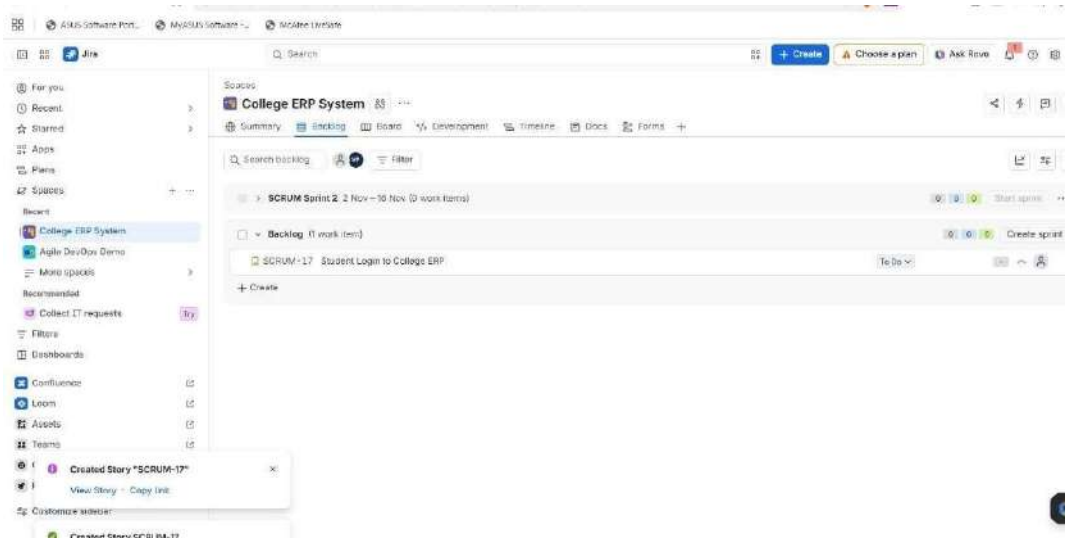
Sprint Review. The Increment was demonstrated to the (simulated) Product Owner. It shows an invalid login rejected with an error message, a successful login with the demonstration credentials, and the Student Dashboard with subject-wise attendance (overall 81.1%) and internal-assessment results for three subjects. All values are illustrative sample data.

Retrospective. What went well: user stories with clear acceptance criteria and sub-tasks made the work visible on the Jira board, and the increment covers login, dashboard, attendance and results. What to improve: the password rules and error messages were unclear at the first Daily Scrum, and the prototype checks credentials in browser JavaScript, which is suitable only for a demonstration. Action items: agree validation rules during Sprint Planning and use server-side authentication with persistent data in the next sprint.

Scrum artifact	**Where it is maintained**
Product Backlog	Jira backlog: SCRUM-17 to SCRUM-20 (Figures 1 to 5)
Sprint Backlog	SCRUM Sprint 2 board with sub-tasks SCRUM-21 to SCRUM-24 (Figures 6 to 11)
Increment	Level-Up-Assignment/Q1-Scrum-Jira/Increment/index.html (Figures 12 to 14)

Output:-

Figure 1: Jira backlog: user story SCRUM-17 (Student Login to College ERP) created



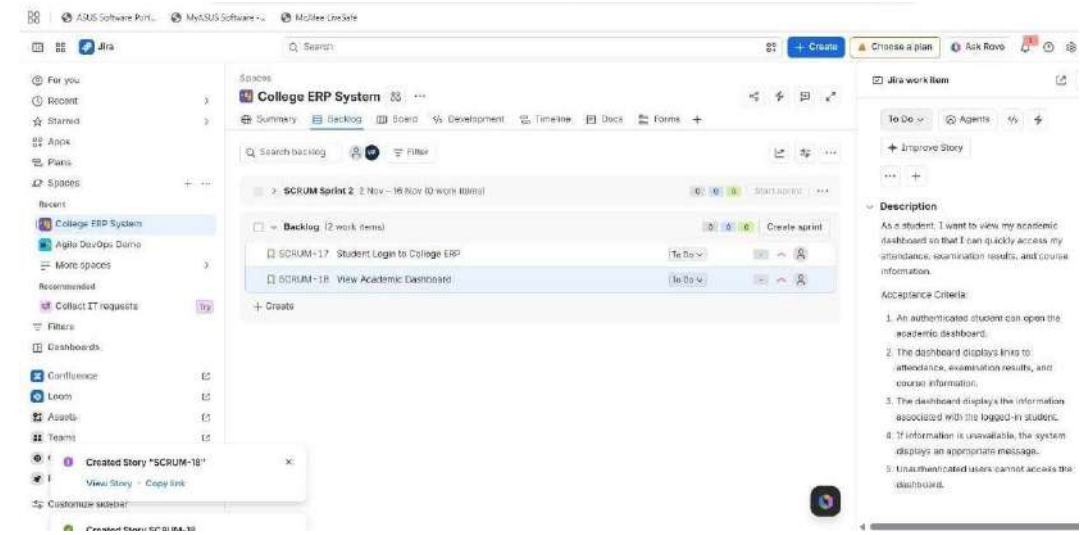
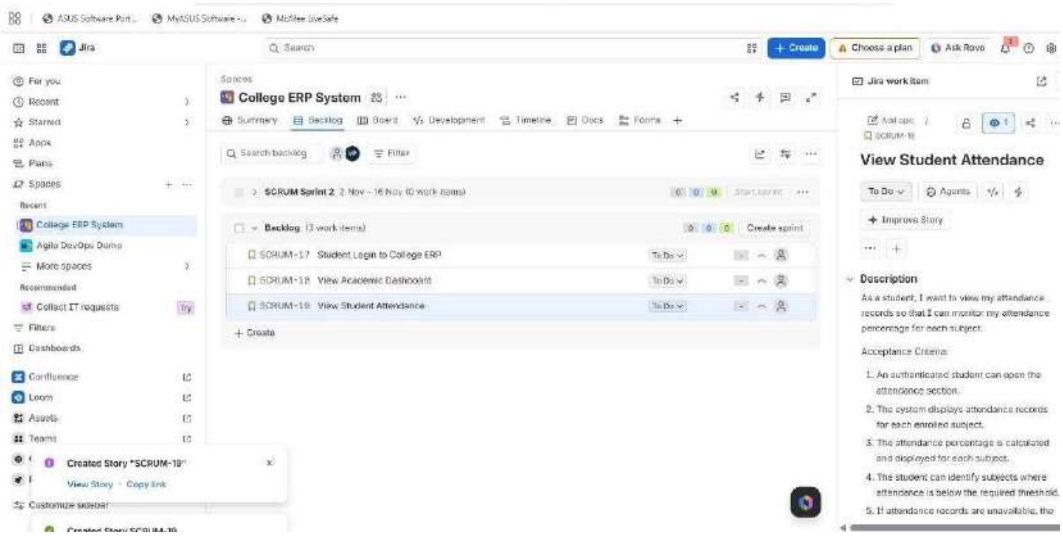


Figure 2: User story SCRUM-18 (View Academic Dashboard) with its acceptance criteria

Figure 3: User story SCRUM-19 (View Student Attendance) with its acceptance criteria



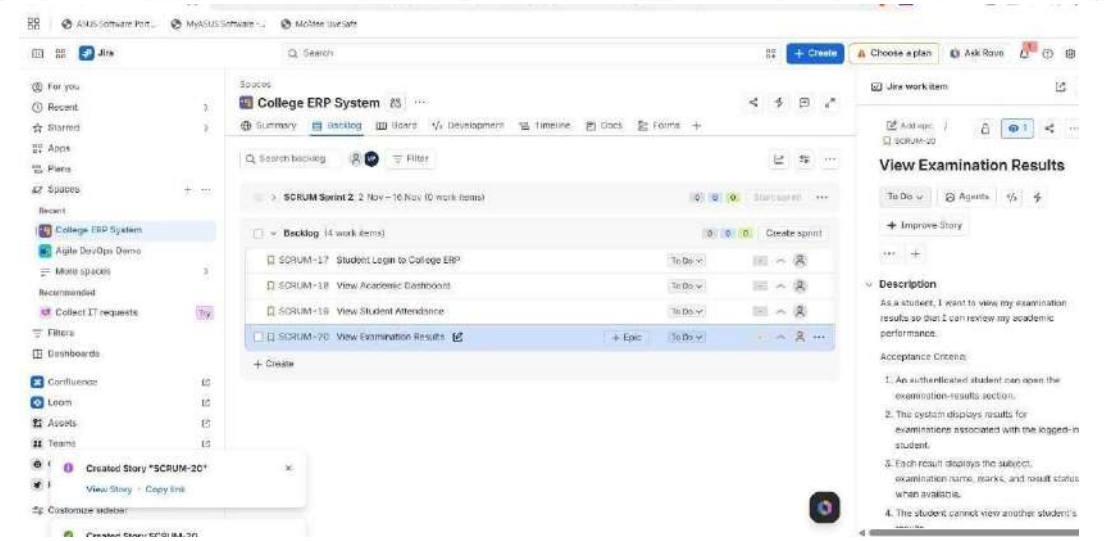
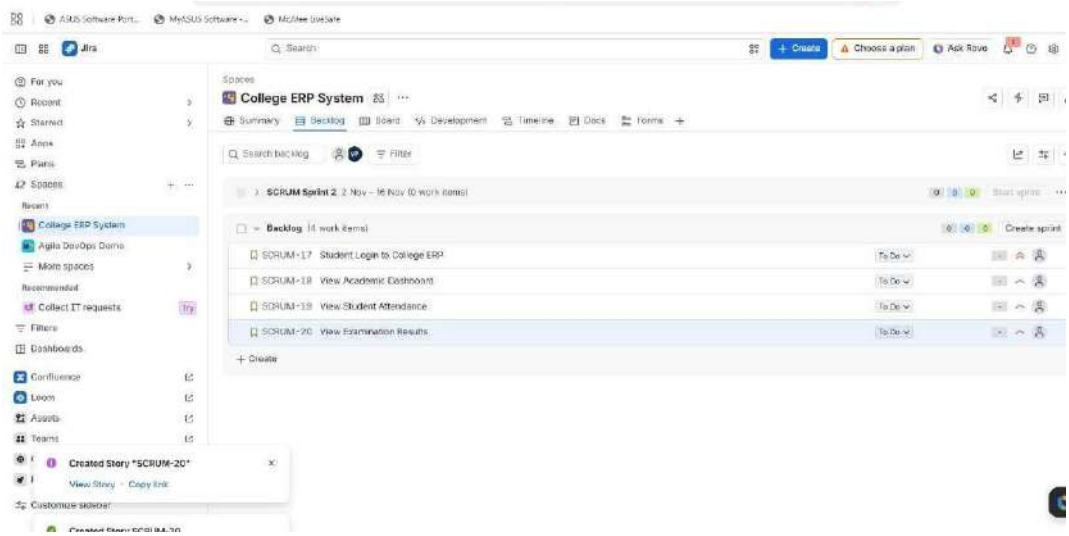


Figure 4: User story SCRUM-20 (View Examination Results) with its acceptance criteria

Figure 5: Prioritised Product Backlog with four stories and SCRUM Sprint 2 (2 Nov – 16 Nov) created



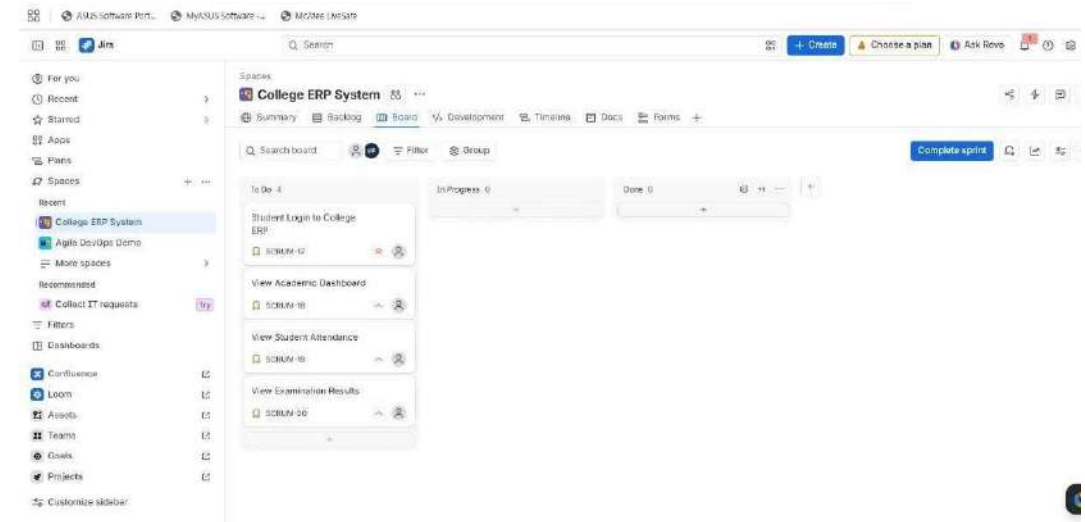
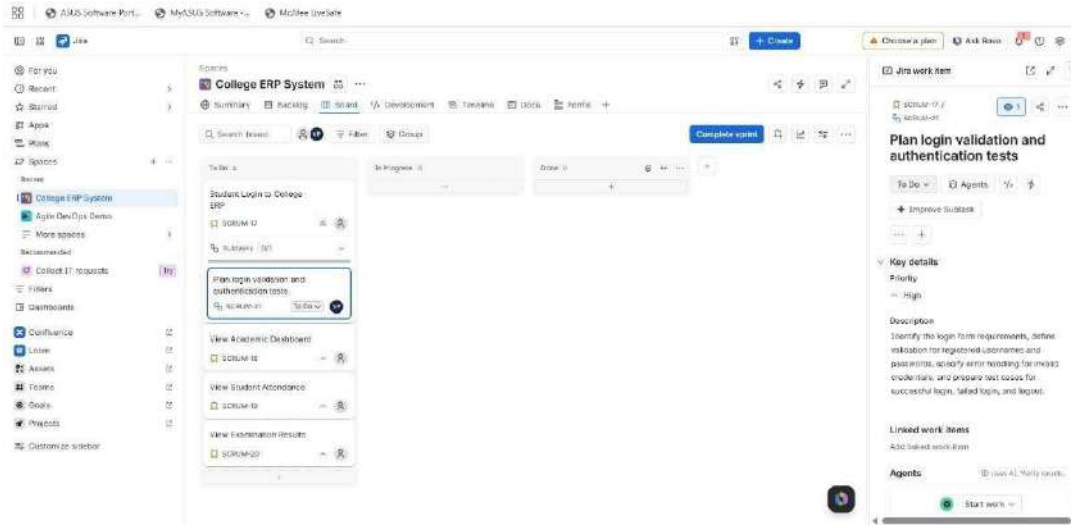


Figure 6: Sprint 2 started: Scrum Board with the four stories in To Do

Figure 7: Sprint Backlog: sub-task SCRUM-21 under SCRUM-17



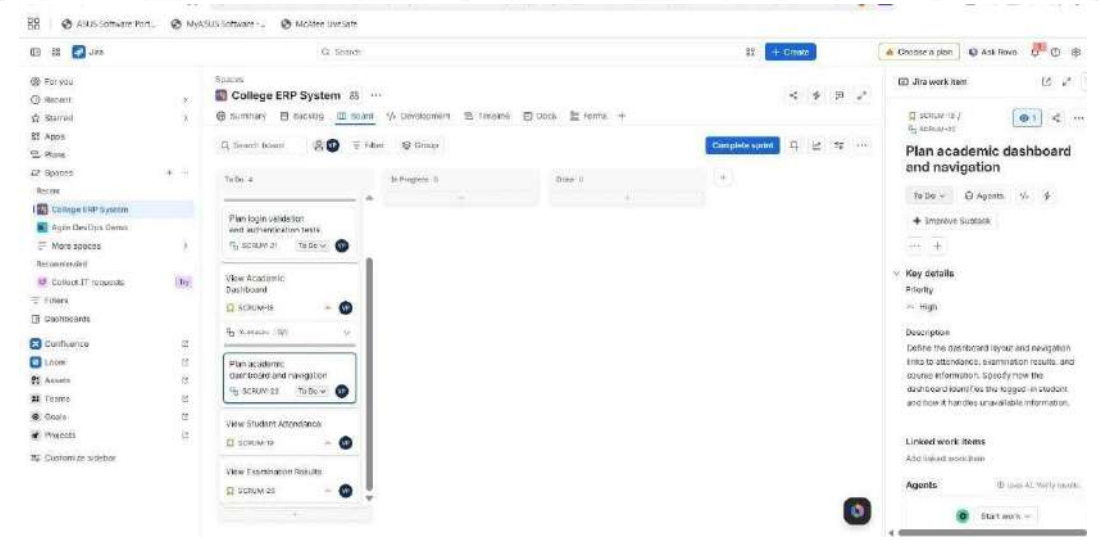
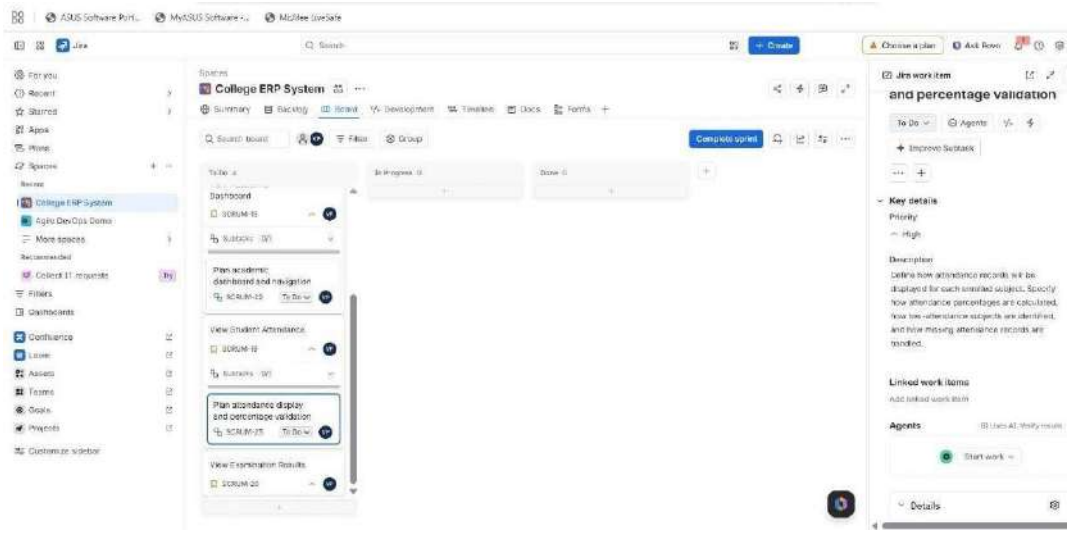


Figure 8: Sprint Backlog: sub-task SCRUM-22 under SCRUM-18

Figure 9: Sprint Backlog: sub-task SCRUM-23 under SCRUM-19



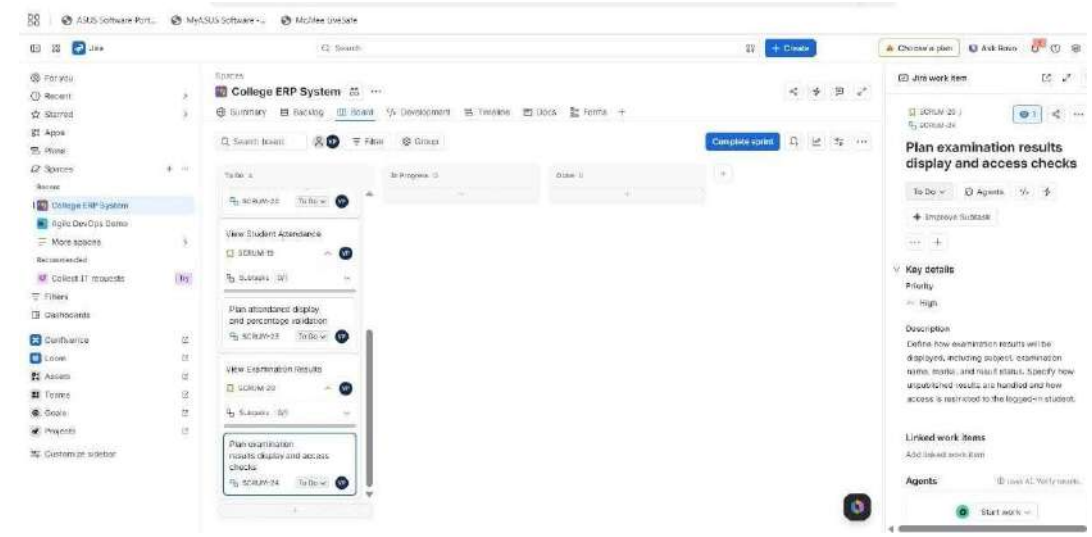
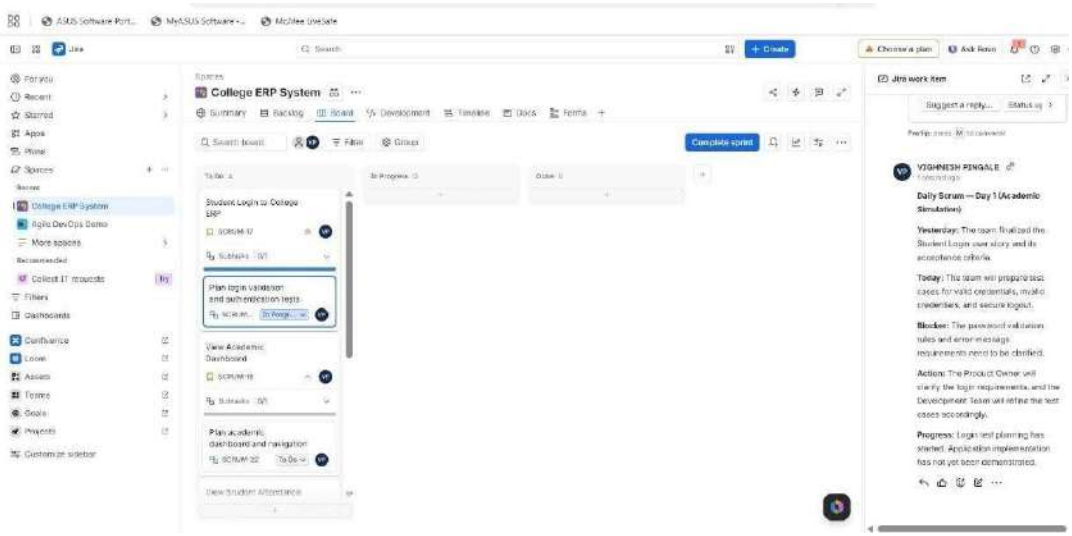


Figure 10: Sprint Backlog: sub-task SCRUM-24 under SCRUM-20 on the Scrum Board

Figure 11: Daily Scrum (Day 1) recorded in Jira; sub-task SCRUM-21 moved to In Progress



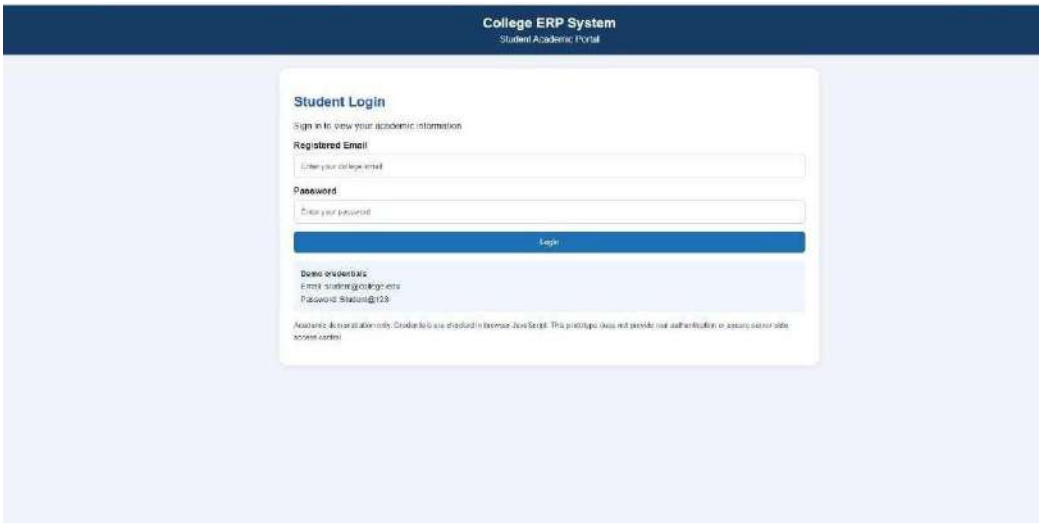
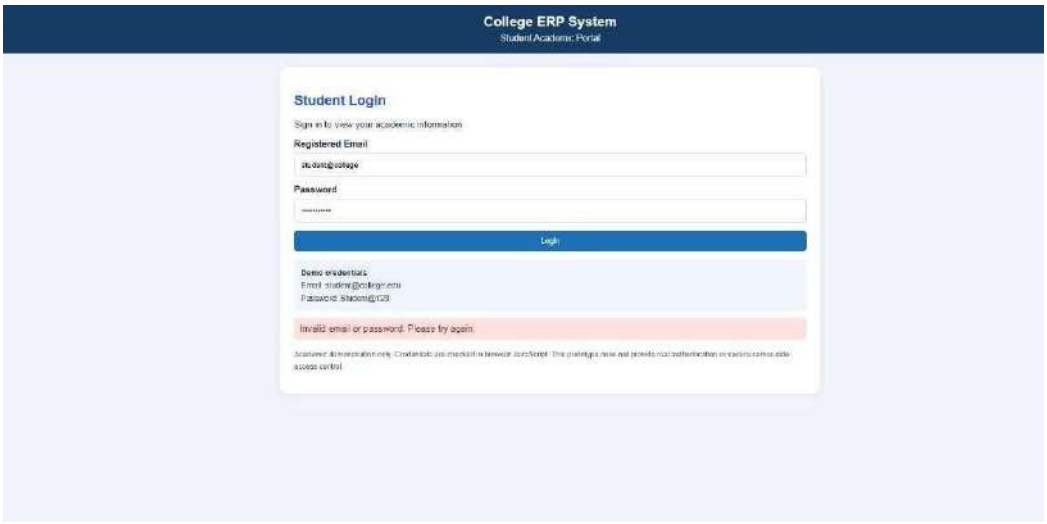


Figure 12: Increment: Student Login page of the College ERP System

Figure 13: Increment: invalid credentials rejected with an error message



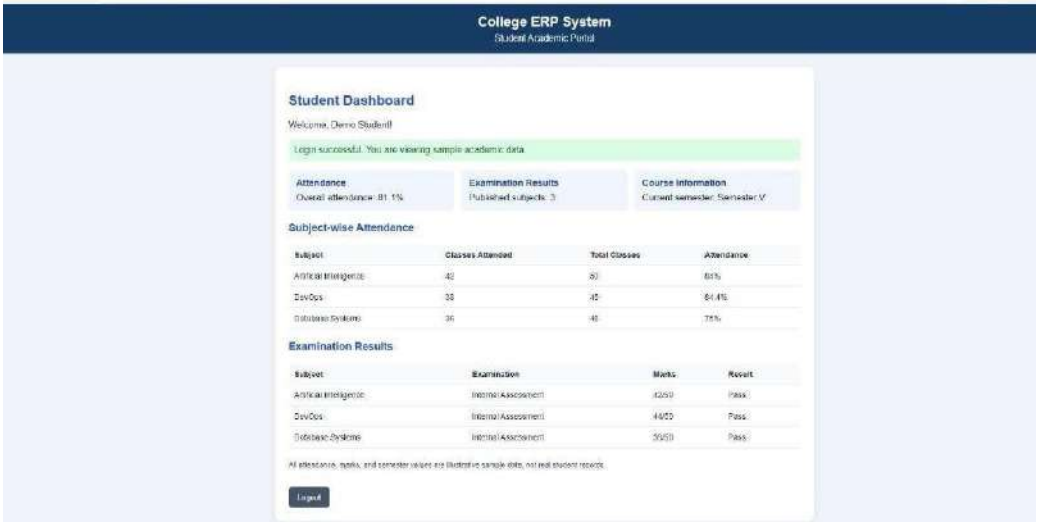


Figure 14: Increment: Student Dashboard with attendance and examination results after login

Q2 : Select a software development scenario such as an Online Food Ordering System, Student Attendance System, or E-Commerce Application. Choose any three Agile development models from the following: Extreme Programming (XP), Feature Driven Development (FDD), Crystal, Kanban, Adaptive Software Development (ASD), Test Driven Development (TDD), Lean Software Development. For each selected model, perform the following activities:

1. Identify the key practices and principles applicable to the selected project.
2. Prepare a workflow or process diagram representing the implementation of the model.
3. Demonstrate the selected model's practices through suitable project activities.
4. Implement at least one practical activity, such as pair programming, test-first development, feature-based planning, Kanban board configuration, or waste identification.
5. Document the outcome with screenshots, sample outputs, or a demonstration of the implemented workflow. [L3]

Implementation:

The scenario is the College ERP student portal (login, attendance and examination results). Three Agile models were selected: Extreme Programming (XP), Kanban and Test Driven Development (TDD). For each model a small browser-based practical was built in Level-Up-Assignment/Q2-Agile-Models and its workflow is shown with a diagram and screenshots.

Model 1: Extreme Programming (XP)

Key practices and principles applied:

- **Pair programming:** the driver implements the change while the navigator reviews the logic and test cases.
- **Small releases:** the password show/hide option and login validation are added as small changes.
- **Continuous testing:** valid and invalid credentials are checked after every change.
- **Feedback:** an immediate, understandable message is shown after each login attempt.
- **Simple design:** the demonstration is kept small and focused on login behaviour.

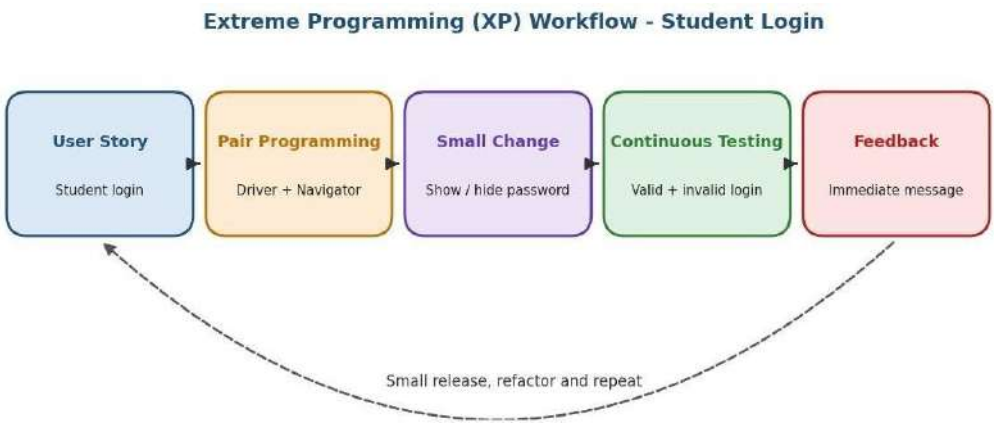
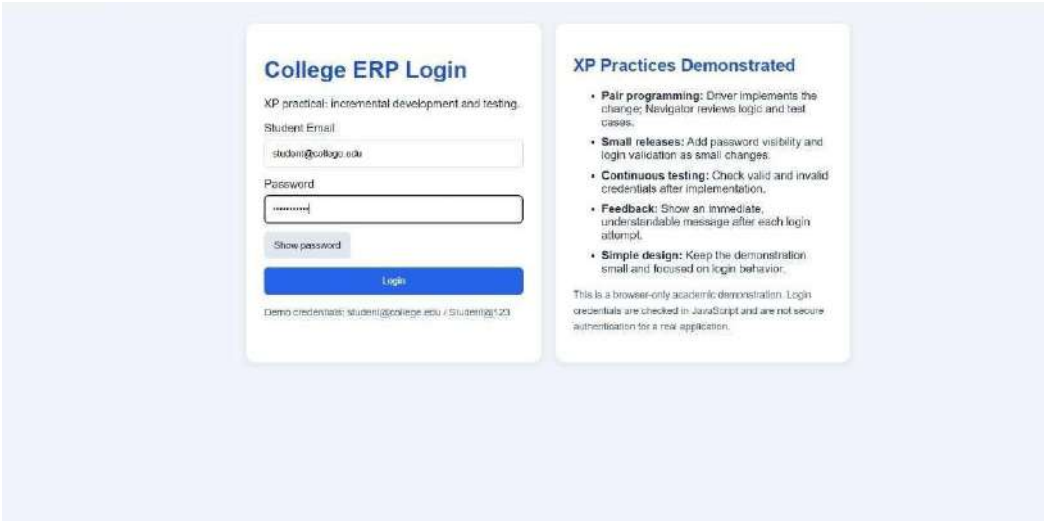


Figure 15: XP workflow applied to the student login feature

Practical activity: pair programming on the login page (xp-pair-programming.html). The driver/navigator pair added a Show password / Hide password button and validated the login. Test inputs: a valid login is accepted, and an incorrect password is rejected with a clear error message.

Figure 16: XP practical: password entered and masked, with the Show password button



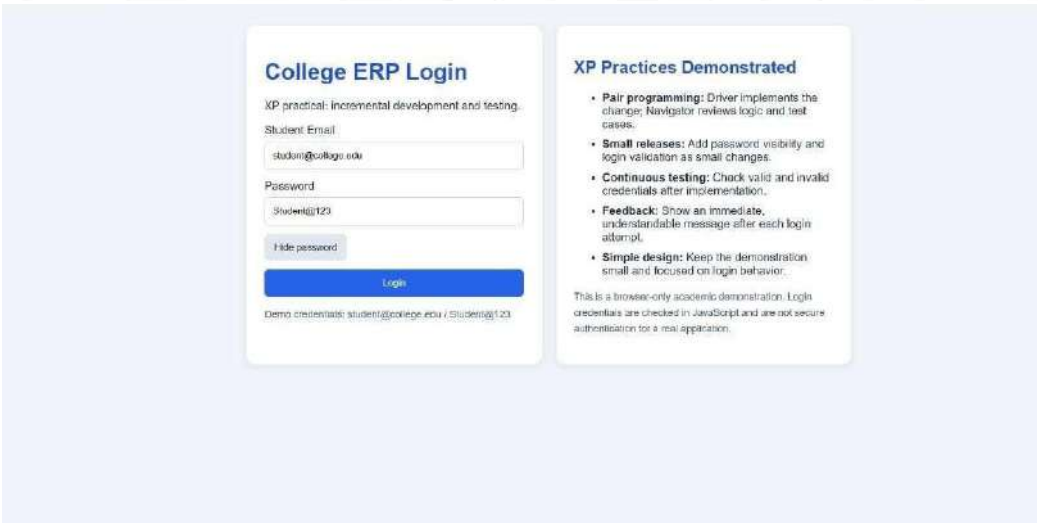
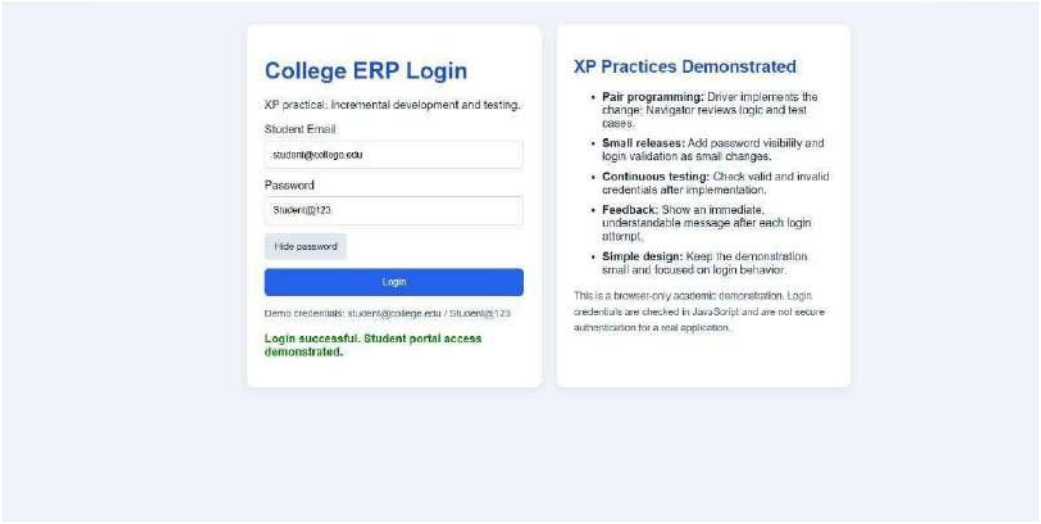


Figure 17: XP practical: Show password toggled (button changes to Hide password)

Figure 18: XP practical: valid credentials give "Login successful. Student portal access demonstrated."



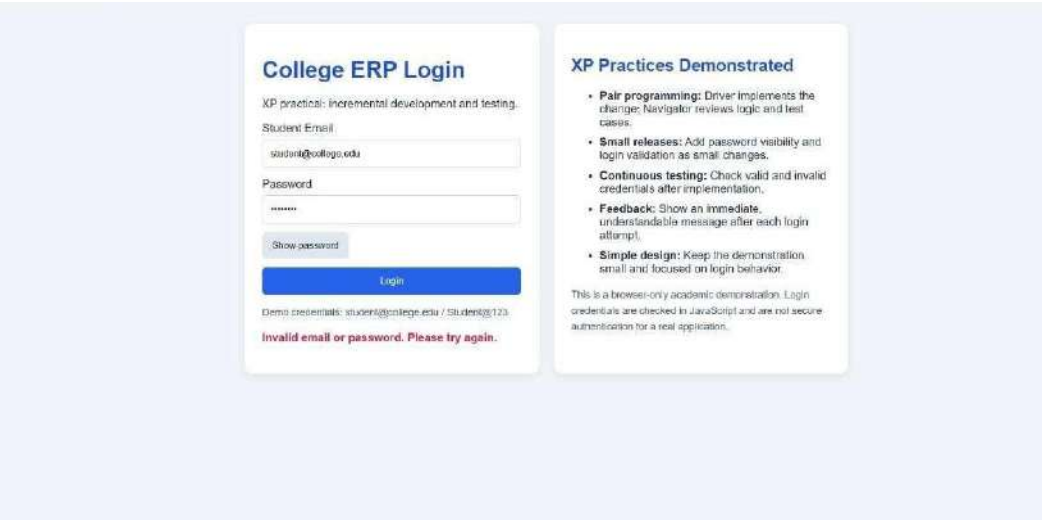


Figure 19: XP practical: invalid credentials give "Invalid email or password. Please try again."

Outcome: the small change was developed, tested with valid and invalid input and gave immediate feedback, which is the XP cycle of pair programming, small release, continuous testing and feedback.

Model 2: Kanban

Key practices and principles applied:

- **Visualise the workflow:** tasks are shown in the columns To Do, In Progress and Done.
- **Limit work in progress:** the In Progress column has a WIP limit of 2 tasks.
- **Pull system and continuous flow:** a task is started (pulled) only when capacity is available and is moved to Done when completed.
- **Manage flow:** tasks can be returned to To Do or reopened from Done.

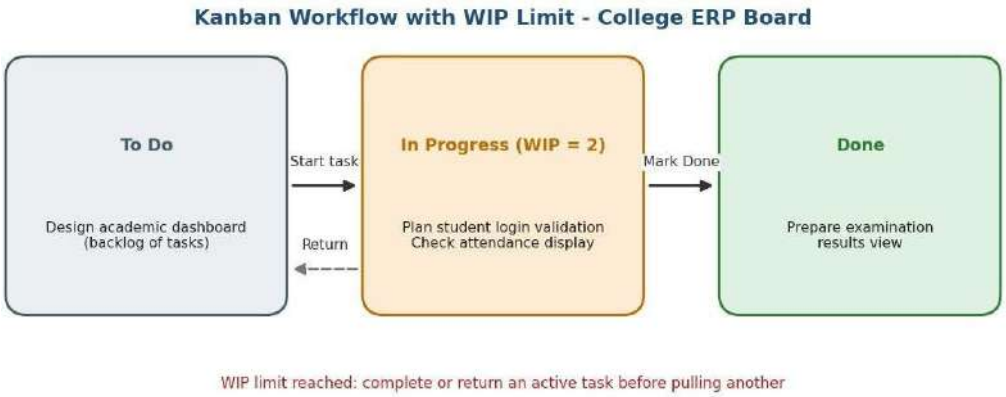


Figure 20: Kanban workflow with a WIP limit of 2 for the College ERP board

Practical activity: Kanban board configuration (kanban-board.html) with the tasks Design academic dashboard, Plan student login validation, Check attendance percentage display and Prepare examination results view. Tasks can be added, started, marked done, returned or reopened, and the board enforces the WIP limit.

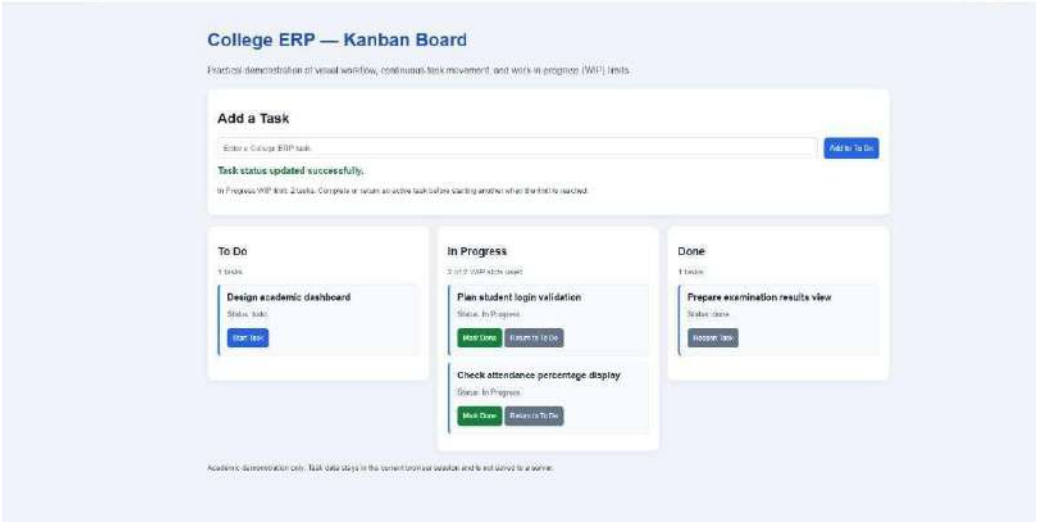
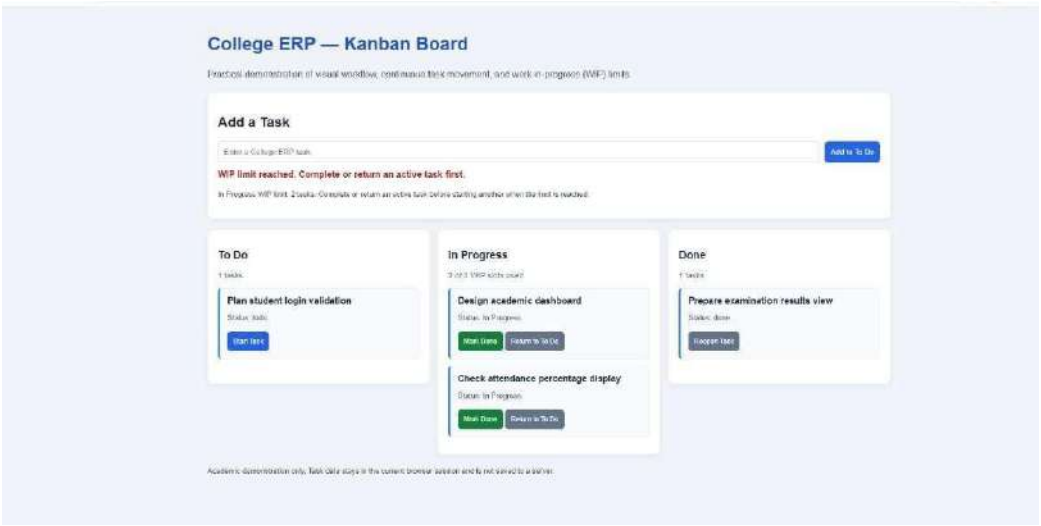


Figure 21: Kanban board: 1 task To Do, In Progress at its WIP limit (2 of 2) and 1 task Done

Figure 22: Kanban board: the WIP limit is enforced ("WIP limit reached. Complete or return an active task first.")



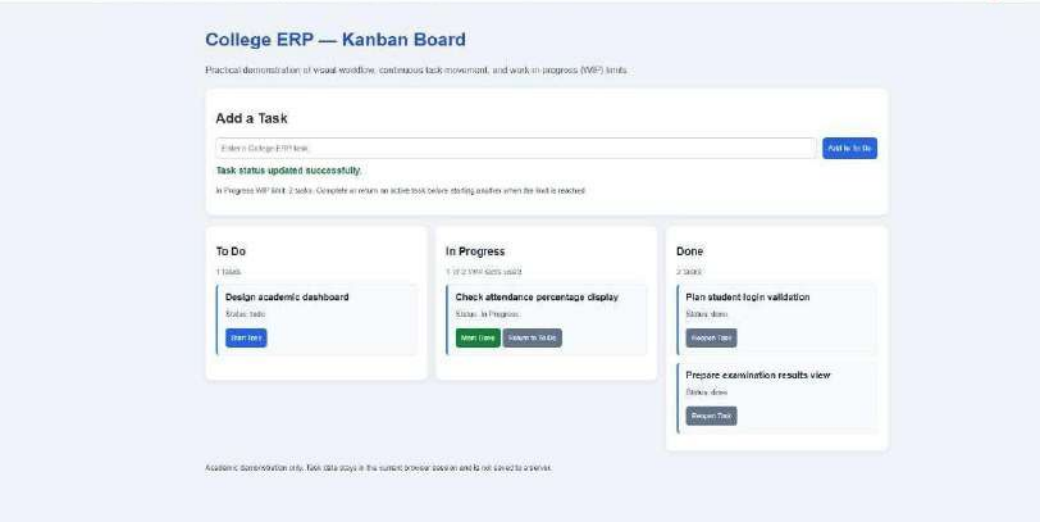


Figure 23: Kanban board after completing a task: 1 of 2 WIP slots used and 2 tasks in Done

Outcome: the board makes the status of every task visible, prevents overloading the team by blocking new work at the WIP limit, and shows how completed work flows to Done.

Model 3: Test Driven Development (TDD)

Key practices and principles applied:

- **Test first:** four automated tests are written before the login validation exists.
- **Red-Green-Refactor:** tests fail first, the smallest implementation makes them pass, and the code is then improved.
- **Fast automated feedback:** the test results are displayed and can be re-run with one click.

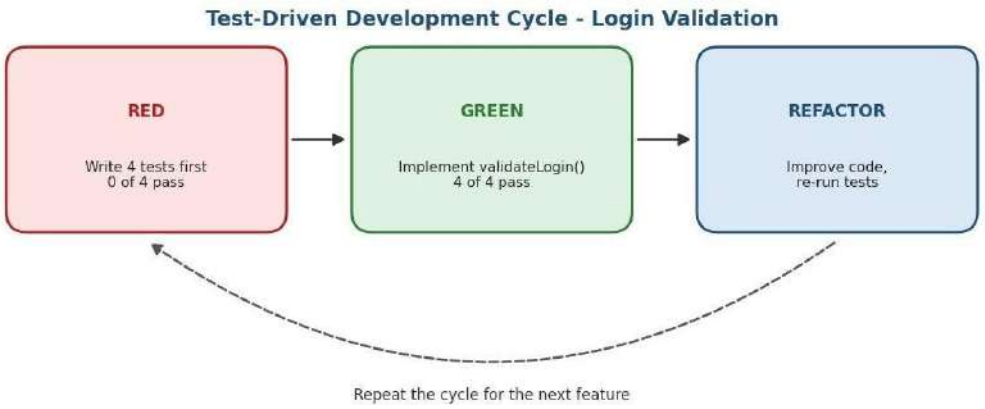


Figure 24: TDD cycle applied to the login validation function

Practical activity: test-first development of validateLogin() (tdd-login-tests.html). The four tests are: a valid email and password should be accepted, an incorrect password should be rejected, an unknown email should be rejected, and empty credentials should be rejected.

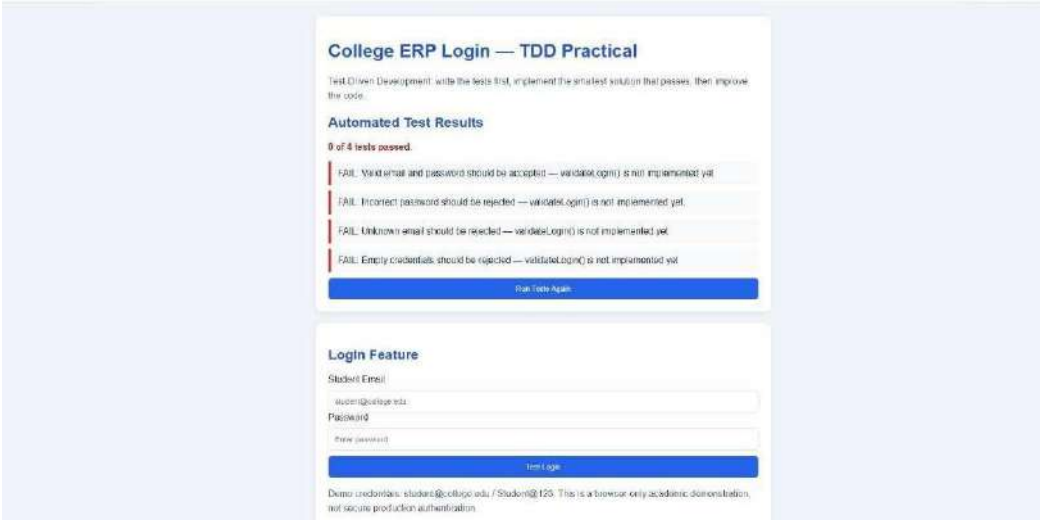
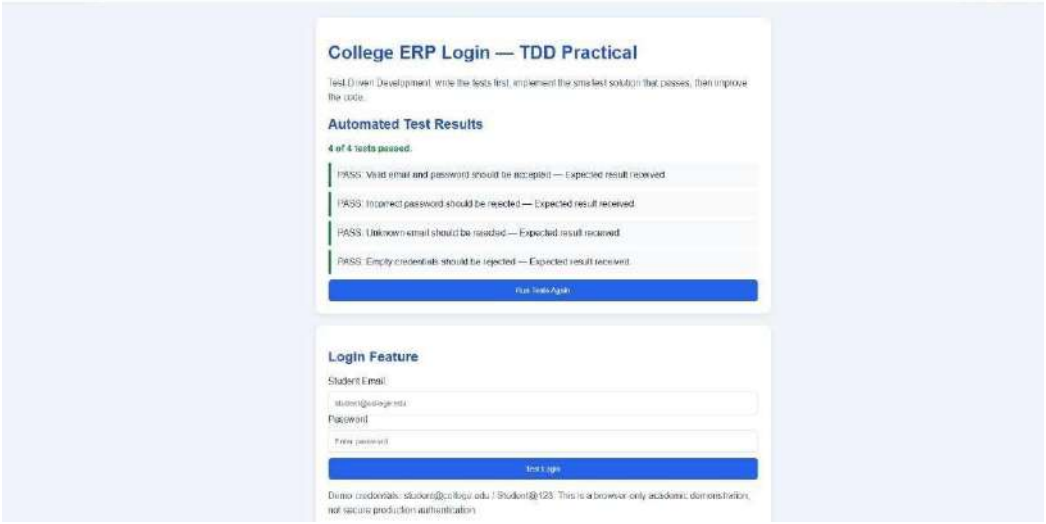


Figure 25: TDD Red stage: 0 of 4 tests pass because validateLogin() is not implemented yet

Figure 26: TDD Green stage: after implementing validateLogin(), 4 of 4 tests pass



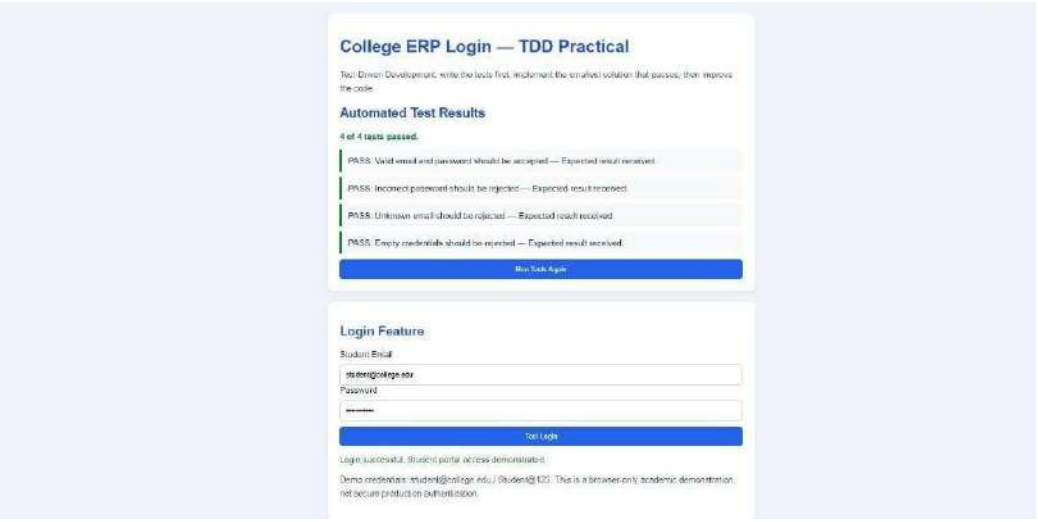


Figure 27: TDD result: the login feature accepts valid credentials while all 4 tests pass

Outcome: the failing-then-passing tests (0 of 4 to 4 of 4) prove that the implementation satisfies the specified behaviour.

Q3 : Consider a software development project in which customer requirements change frequently, development resources are limited, and stakeholders provide continuous feedback. Select a suitable project scenario and perform the following activities:

1. Prepare an initial set of project requirements and create a Product Backlog.
2. Simulate a change in customer requirements during development and update the Product Backlog accordingly.
3. Prioritize requirements based on business value, urgency, and available resources.
4. Plan a short development iteration by assigning tasks to available team members.
5. Demonstrate stakeholder collaboration through a simulated customer feedback or review session.
6. Implement changes based on feedback and demonstrate the updated working Increment.
7. Prepare a brief report explaining how the applied Agile practices helped manage changing requirements, resource constraints, and customer expectations. [L3]

Implementation:

Scenario: the College ERP team is developing a student portal. Requirements may change after stakeholder feedback, and the team has limited time and development capacity, so features are prioritised by business value, urgency and available resources. The backlog, report and increment are stored in Level-Up-Assignment/Q3-Changing-Requirements.

1. Initial requirements and Product Backlog

1. Students must log in using valid credentials.
2. Students must view an academic dashboard.
3. Students must view subject-wise attendance percentages.
4. Students must view their examination results.

At this stage the dashboard has no dedicated warning for attendance below 75%. The requirements were recorded as user stories REQ-01 and REQ-02 (login, 5 points, and dashboard, 3 points, both High business value and Must priority) and the further stories in the backlog document.

2. Change in requirements and updated Product Backlog

During the simulated review the stakeholder requested a new requirement, REQ-05: add a low-attendance warning to the academic dashboard. The Product Backlog was updated with REQ-05 and its status.

3. Prioritisation

REQ-05 was given priority because it has high business value (students must identify attendance risks before examinations) and is urgent for the same reason, while it needs only a small change to the existing dashboard, which fits the limited development and testing capacity of the team.

4. Iteration plan and task assignment

A short iteration of 3 working days was planned with two simulated team members:

- **Developer:** add the low-attendance warning to the dashboard and verify that normal attendance does not show a warning.
- **Tester / reviewer:** test attendance below 75%, equal to 75% and above 75%, record the results and report defects.

Acceptance criteria for REQ-05: (1) a warning appears when attendance is below 75%; (2) no low-attendance warning appears at exactly 75%; (3) no warning appears above 75%; (4) the displayed attendance percentage remains visible; (5) the warning can be understood without relying on colour alone.

5. Stakeholder collaboration

Simulated customer review. The college representative (simulated) said that the portal should clearly warn students when their attendance in any subject falls below 75%, because students need to identify attendance risks before examinations. The requested change was to add a low-attendance warning to the academic dashboard, since students may overlook low attendance when only a percentage is displayed.

6. Implemented change and updated Increment

The warning was implemented in the browser-based prototype and verified with three sample attendance values:

Test	Sample attendance	Observed result
Q3-T1	68%	Warning displayed
Q3-T2	75%	No low-attendance warning
Q3-T3	82%	No low-attendance warning

Output:-

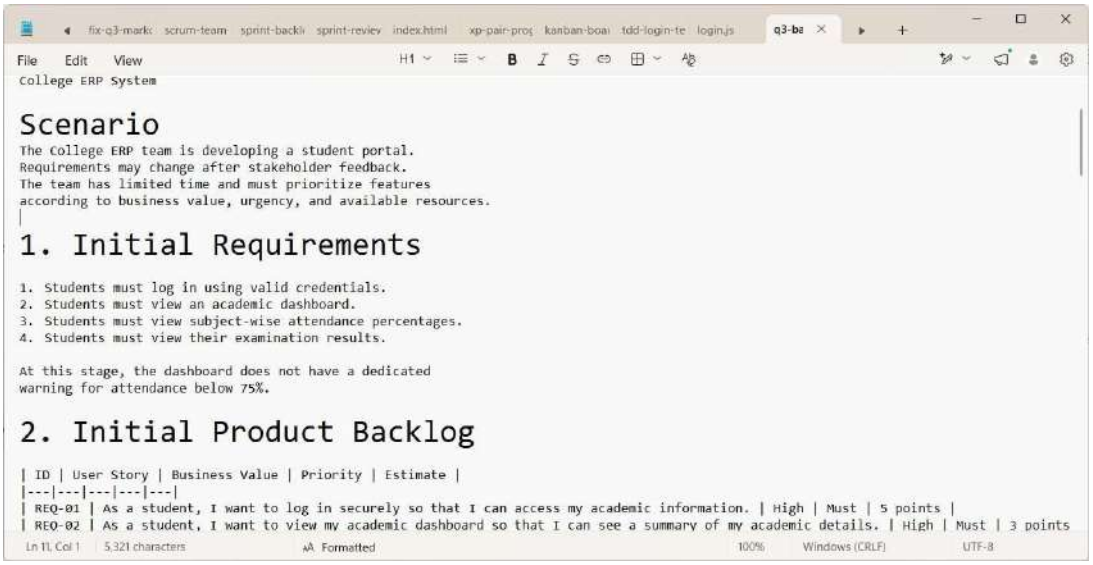


Figure 28: Q3 backlog document: scenario, initial requirements and initial Product Backlog

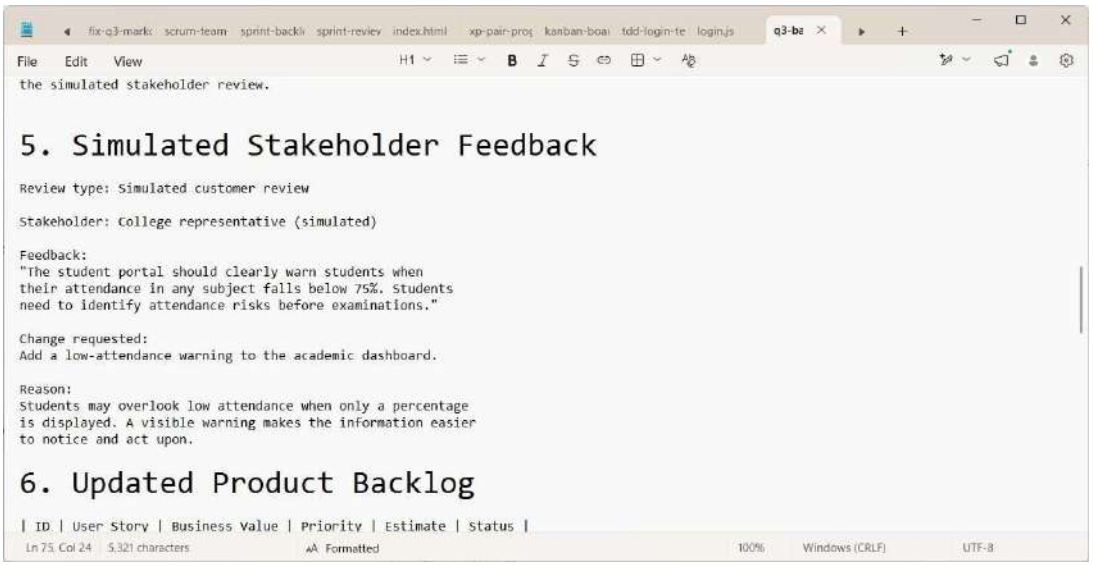


Figure 29: Simulated stakeholder feedback and the requested change

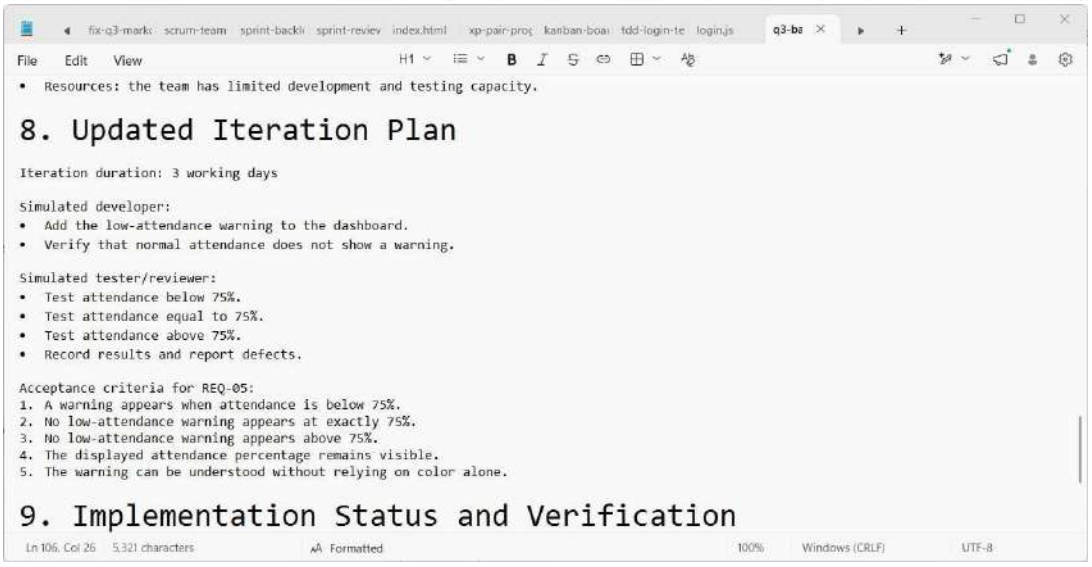


Figure 30: Updated iteration plan (3 working days) and acceptance criteria for REQ-05

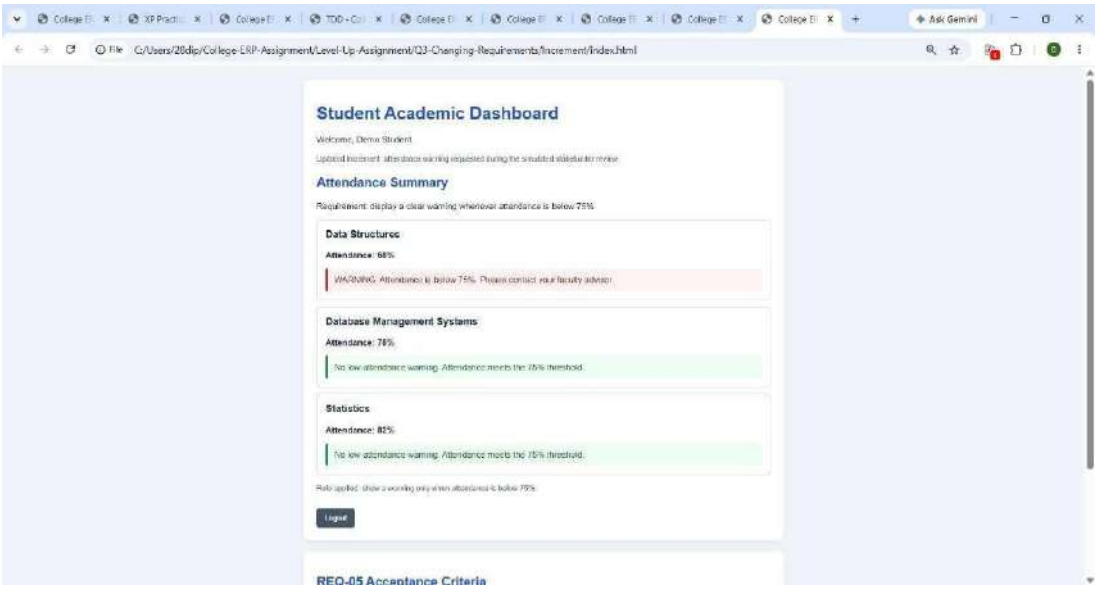


Figure 31: Updated Increment: Data Structures (68%) shows the warning; 75% and 82% show none

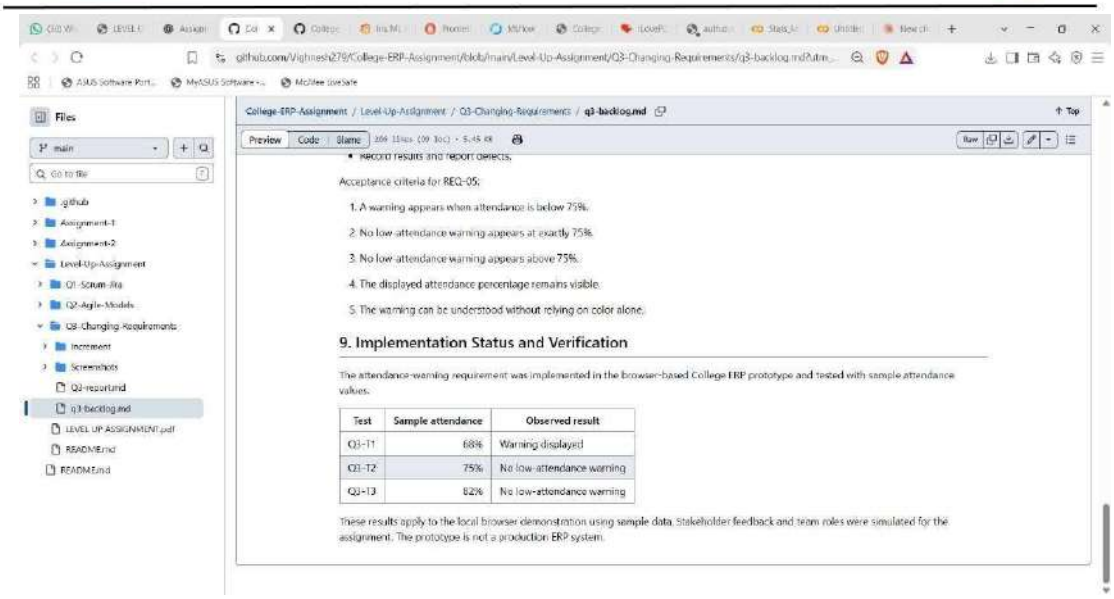


Figure 32: Implementation status and verification table in the repository (Q3-T1 to Q3-T3)

7. Brief report

Changing requirements. Because the work was held in a Product Backlog, the new low-attendance requirement could be added, re-prioritised and planned for the next short iteration without redoing earlier work.

Resource constraints. Prioritising by business value, urgency and effort, and planning a 3-day iteration with a clear developer and tester split, let the limited team deliver the most valuable change first.

Customer expectations. The stakeholder review and the acceptance criteria made the expectation measurable, and the verification at 68%, 75% and 82% showed that the delivered Increment matches it. The attendance percentage stays visible and the warning is text-based, so it does not depend on colour alone.

Conclusion:

The Level Up Assignment applies Scrum with Jira (Q1), three Agile models – XP, Kanban and TDD – with working practicals (Q2), and Agile practices for changing requirements, limited resources and continuous stakeholder feedback (Q3), all on the College ERP System. Stakeholder feedback and team roles were simulated for the assignment, and the prototypes use sample data.

GitHub repository: https://github.com/Vighnesh279/College-ERP-Assignment